

Matrices

4.13 Spectral Decomposition Law

Once the eigenvalues and eigenvectors of a Hermitian matrix H have been found, the expression we now derive for H is referred to as its spectral decomposition.

$$H = \sum_i f(\lambda_i) |r_i\rangle \langle r_i|$$

Each r_i satisfies $Hr_i = f(\lambda_i)r_i$ and $\langle r_i | r_i \rangle = 1$.

Another result related to the spectral decomposition of H can be obtained if we multiply both sides of the equation $Hr_i = \lambda_i r_i$ on the left by H , reaching

$$H^2 r_i = \lambda_i^2 r_i$$

Further applications of H show that all positive powers of H have the same eigenvectors as H , so if $f(H)$ is any function of H that has a power-series expansion, it has the spectral decomposition $f(H) = \sum_i f(\lambda_i) |r_i\rangle\langle r_i|$

The above equation can be extended to include negative powers if H is nonsingular; to do so,

multiply $Hr_i = \lambda_i r_i$ on the left by H^{-1} and rearrange, to obtain $H^{-1}r_i = \frac{1}{\lambda_i} r_i$

showing that negative powers of H also have the same eigenvectors as H .

Example: $A_{3 \times 3} \Rightarrow |r_1\rangle_{3 \times 1} \langle r_1|_{1 \times 3}$

$$f(A_{3 \times 3}) = f(\lambda_1) |r_1\rangle\langle r_1| + f(\lambda_2) |r_2\rangle\langle r_2| + f(\lambda_3) |r_3\rangle\langle r_3|$$

Question: $\sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, Then find e^{σ_3} .

Solution: Eigenvalues : $(1 - \lambda)(-1 - \lambda) = 0$

$$\lambda = \pm 1, \lambda_1 = 1, \lambda_2 = -1$$

Eigenvector:

$$\lambda_1 = 1 \rightarrow r_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \lambda_2 = -1 \rightarrow r_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$e^{\sigma_3} = e^1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + e^{-1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$e^{\sigma_3} = \begin{bmatrix} e & 0 \\ 0 & \frac{1}{e} \end{bmatrix}$$

Example: Consider a matrix $M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

(a) Eigenvalues

(b) $e^M = ?$

Solution: (a) $\begin{bmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$(1-\lambda)\left[(1-\lambda)^2-1\right]-1\left[(1-\lambda)-1\right]+1[1-1+\lambda]=0$$

$$\lambda_1=0, \lambda_2=0, \lambda_3=3$$

$$(b) e^M = I + \frac{M}{1!} + \frac{M^2}{2!} + \dots$$

$$M^2 = 3M, M^3 = 3^2 M, M^4 = 3^3 M$$

$$e^M = I + M + \frac{3M}{2!} + \frac{3^2 M}{3!} + \dots = I + M \left[1 + \frac{3}{2!} + \frac{3^2}{3!} + \dots \right]$$

$$e^M = I + M \left[\frac{1}{3} \left(1 + \frac{3}{1!} + \frac{3^2}{2!} + \dots \right) - \frac{1}{3} \right]$$

$$e^M = I + \left(\frac{e^3 - 1}{3} \right) M$$

Question: A 3×3 matrix M has $Tr(M) = 6, Tr(M^2) = 26, Tr(M^3) = 90$

Which of the following options can be the possible set of eigenvalues?

- (a) 1, 1, 4 (b) -1, 3, 4 (c) -1, 0, 7 (d) 2, 2, 2

Answer: (b)

Solution: $T_r[M] = -1 + 3 + 4 = 6$

$$T_r[M^2] = 1^2 + 3^2 + 4^2 = 16 + 9 + 1 = 26$$

$$T_r[M^3] = (-1)^3 + 3^3 + 4^3 = -1 + 27 + 64 = 90$$

Question: The eigenvalues of anti-symmetric matrix

$$A = \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix}$$

Where n_1, n_2, n_3 are the components of a unit vector. Then its eigenvalues will be:

- (a) $0, -i, i$ (b) $0, 1, -1$ (c) $0, 1, +i, 1-i$ (d) $0, 0, 0$

Answer: (a)

Solution: $|A - \lambda I| = \begin{vmatrix} 0 - \lambda & -n_3 & n_2 \\ n_3 & 0 - \lambda & -n_1 \\ -n_2 & n_1 & 0 - \lambda \end{vmatrix}$

$$= (-\lambda) [\lambda^2 + n_1^2] + n_3 [-\lambda n_3 - n_1 n_2] + n_2 [n_1 n_3 - \lambda n_2] = 0$$

$$= -\lambda^3 - \lambda n_1^2 - \lambda n_3^2 - n_1 n_2 n_3 + n_1 n_2 n_3 - \lambda n_2^2 = 0$$

$$= -\lambda^3 - \lambda [n_1^2 + n_2^2 + n_3^2] = 0$$

$$-\lambda [\lambda^2 + 1] = 0 \quad \lambda = 0, \pm i$$

And also since the matrix is antisymmetric so eigenvalues should be either zero or pure imaginary.

Question: Given a 2×2 Unitary matrix U satisfying $U^\dagger U = I = U U^\dagger$, with determinant $U = e^{i\phi}$, construct a unitary matrix V ($V^\dagger V = V V^\dagger = I$) with determinant $V = 1$ from it by

- (a) Multiplying U by $e^{-i\phi/2}$
- (b) Multiplying any single element of U by $e^{-i\phi}$
- (c) Multiplying any row or column of U by $e^{-i\phi/2}$
- (d) Multiplying U by $e^{-i\phi}$

Answer: (a)

Solution: $\det(\alpha U) = \alpha^n \det U$, $n = 2$

$$1 = \alpha^2 e^{i\phi} \Rightarrow \alpha^2 = e^{-i\phi}, \quad \alpha = e^{-i\phi/2}$$

Question: Consider the matrix equation

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & b & 2c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The condition for the existence of a non-trivial solution and the corresponding normalized solution is.

Solution: $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & b & 2c \end{bmatrix}$

$$[4c-3b]-1[2c-6]+1[b-4]=0$$

$$4c-3b-2c+6+b-4=0$$

$$2c-2b+2=0 \quad 2c=2b-2 \Rightarrow b=c+1$$

Question: Consider the three vectors $\vec{v}_1 = 2\hat{i} + 3\hat{k}$, $\vec{v}_2 = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{v}_3 = 5\hat{i} + \hat{j} + a\hat{k}$. These vectors will be linearly dependent if the value of a is:

$$\text{Solution: } [\vec{v}_1 \quad \vec{v}_2 \quad \vec{v}_3] = \begin{bmatrix} 2 & 1 & 5 \\ 0 & 2 & 1 \\ 3 & 2 & a \end{bmatrix} \Rightarrow 2(2a-2)-1(-3)+5(-6)=0$$

$$4a-4+3-30=0 \quad 4a-1-30=0$$

$$4a-31=0 \quad a = \frac{31}{4}$$

Special Unitary Matrix:

$$SU(2) = \begin{Bmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{Bmatrix}$$

$$|\alpha|^2 + |\beta|^2 = 1$$

$$\text{Example: } \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$$

$$\Rightarrow \lambda^2 - \lambda(Tr(A)) + \det A = 0$$

$$\lambda^3 - Tr(A)\lambda^2 + (a_{11} + a_{22} + a_{33})\lambda - \det A = 0.$$